Introduction

The aim of the project is to gain an **overview of how data science works**, learn the fundamental concepts and techniques of data analysis, understand how they work, and develop intuition for their appropriate application in order to discover knowledge from data. Students will also gain an understanding of what kinds of questions can be answered using data analysis and how to apply **fundamental machine learning approaches**. Emphasis is placed on data analysis and preprocessing, the application of machine learning methods, and ways of evaluating and comparing them.

The project is completed **in pairs** and must meet an acceptable level of quality. **Python** and available data science libraries such as pandas, NumPy, SciPy, statsmodels, scikit-learn, etc. are used. For each phase and activity, an executable **Jupyter Notebook** must be submitted to UIS, containing all transformations performed on the data together with appropriate documentation. The submitted notebook must contain not only the code but also its outputs (calculated values, printed results, visualizations, etc.), along with comments on the obtained results and the resulting decisions regarding the next steps in the data analysis process. The ability to effectively communicate and present relevant results constitutes an important part of the assessment.

For each phase, the submitted notebook must state **the percentage contribution** of each member of the pair.

Data

https://drive.google.com/drive/folders/1qI7dqHpDH7lbdgbB_9i3ZSsoCuhR766F?usp=sharing
(each pair has one dataset corresponding to the number assigned to you during the exercise)

Predictive Maintenance (PdM) is a modern approach to monitoring technical equipment, aimed at the early identification of emerging faults before they fully manifest. Failures occurring in public transport vehicles during regular operation can lead to service interruptions, unplanned downtime, and significant economic losses. In addition to their negative impact on the operator, such situations can also reduce the reliability of public transport services and passenger satisfaction. Traditional approaches based on scheduled maintenance or fixed sensor thresholds often fail to detect emerging faults sufficiently early.

Modern machine learning methods make it possible to analyze continuously collected sensor data and identify anomalies that may indicate an impending equipment failure. Models use data from various sensors monitoring the condition of individual components and can recognize hidden patterns of behavior that may lead to failure. In addition to anomaly detection, these approaches can also support the prediction of future failures, estimation of the remaining useful life of components, and analysis of the causes of failure. This makes it possible to schedule maintenance before critical failures occur, minimize unplanned downtime, and reduce repair costs.

The aim of this assignment is to become familiar with the field of predictive maintenance and to understand the potential of artificial intelligence, particularly machine learning, for detecting critical system failures in sensor data, thereby enabling timely maintenance.

Project assignment

Rail Air-Production-Unit (APU) Predictive Maintenance

The QUEST

Each pair will work with an assigned dataset starting from Week 2. **Your task** is to predict the values of the dependent variable "**failure**" **(target variable)** using machine learning methods. In doing so, you will need to address various issues present in the data, such as data formats, missing values, outliers, and many others.

The expected **output** of the project is:

1. the best machine learning model;
2. a data pipeline for building the model based on the input data.

Phase 1 – Exploratory Data Analysis: 15 points

1.1 Basic Description of the Data and Its Characteristics (5 points)

EDA with visualization

- **(A – 1 point)** Analyze the structure of the data, including files (their structure and relationships, number of files, types, etc.) and records (structure, number of records, number of attributes, data types, etc.).
- **(B – 1 point)** Analyze individual attributes: for selected important attributes (at least 10), analyze their distributions and basic descriptive statistics, and determine whether they meet the specified conditions and ranges of measured values.
- **(C – 1 point)** Pairwise data analysis: identify relationships and dependencies between pairs of attributes.
- **(D – 1 point)** Pairwise data analysis: identify dependencies between the **target variable** and the other variables (potential predictors).
- **(E – 1 point)** Document your initial thoughts on how to approach the project task, e.g., Are some attributes dependent on each other? Which attributes does the target variable depend on? Is it necessary to combine records from multiple files?

1.2 Identification of Problems, Data Integration, and Data Cleaning (5 points)

- **(A – 2 points)** Identify and initially address problems in the data, such as inappropriate data structure, duplicate records that may arise after certain data transformations, inconsistent formats, missing values, and outliers. The data may also contain other problems not listed here, or fewer problems than those mentioned.
- **(B – 2 points)** Check the correctness and consistency of the data:
  - determine whether the data contain abnormal values;
  - determine whether the data contain illogical relationships resulting from the data collection and annotation processes.
- **(C – 1 point)** Outlier detection: address the problem using at least 2 techniques, such as:
  - removing outliers or extreme observations;
  - replacing outlier values with boundary values of the distribution (e.g., 5%, 95%).

1.3 Formulation and Statistical Verification of Hypotheses about Data (5 points)

- **(A – 4 points)** Formulate two hypotheses about the data in the context of the assigned prediction task. Test the formulated hypotheses using appropriately selected statistical tests. Example formulation:

  flowmeter has, on average, a higher value in the state with failure than without it.
- **(B – 1 point)** Verify whether your statistical tests have sufficient support from the data, i.e., whether they have sufficient **statistical power**.

In the submitted report (Jupyter Notebook), you should therefore answer the following questions: Are the data in a suitable format for further processing? What problems are present in the data? Do any attributes contain inconsistent values? How do you address the problems you have identified?
The report is due in Week 5 of the semester. Each pair will present the completed phase in their notebook to their instructor during the practical session, as required. State the percentage contribution of each member of the pair. The report must then be submitted electronically by **one member of the pair** via UIS by Sunday, **18 October 2026 at 23:59**.

Phase 2 – Data Preprocessing: 15 points

In this phase, you are expected to perform data preprocessing **for machine learning**. The result should be a dataset (CSV or TSV) in which each observation is represented by a single row.

- **scikit-learn** can only work with numerical data, so non-numerical data must be appropriately handled.
- the preprocessing must be reproducible for both the training and test datasets, so that you can repeat the preprocessing multiple times as needed (iteratively).

Since preprocessing may change the shape and characteristics of the data, you should repeat Exploratory Data Analysis (EDA) as needed. The EDA techniques will not be graded again. Document any changes to the selected approaches. Data-related problems can be addressed iteratively in each phase and across all phases as needed.

2.1 Data Preprocessing Implementation (5 points)

- **(A – 1 point)** Split the data into training and test sets according to a predefined ratio of your choice. From this point onward, **work only with the training dataset**.
- **(B – 1 point)** Transform the data into a suitable format for machine learning, i.e., each observation must be represented by one row and every attribute must be in numerical format. Iteratively integrate the data preprocessing steps from Phase 1 into the overall preprocessing process.
- **(C – 1 point)** Transform the data attributes for machine learning using the available techniques, including at least: scaling (2 techniques), transformers (2 techniques), other appropriate techniques. The goal is to test their effects and appropriately combine them in a data pipeline (starting from Section 2.3 and continuing in Phase 3).
- **(D – 2 points)** Justify your choices and decisions regarding the implementation (i.e., provide documentation).

2.2 Feature Selection for Machine Learning (5 points)

- **(A – 3 points)** Determine which attributes (features) in your ML dataset are informative with respect to the target/predicted variable. Use at least 3 different techniques and compare their results.
- **(B – 1 point)** Rank the identified features according to their importance.
- **(C – 1 point)** Justify your choices and decisions regarding the implementation (i.e., provide documentation).

2.3 Reproducibility of Preprocessing (5 points)

- **(A – 3 points)** Modify the code used to preprocess the training set so that it can be reused without any additional modifications to preprocess **the test set** in the machine-learning context.
- **(B – 2 points)** Make use of **sklearn.pipeline** functionality.

The report is due in Week 7 of the semester. Each pair will present the completed phase in their notebook to their instructor during the practical session, as required. State the percentage contribution of each member of the pair. The report must then be submitted electronically by **one member of the pair** via UIS by Sunday, **01 November 2026 at 23:59**.

Phase 3 – Machine Learning: 20 points

The report is due in Week 10 of the semester. Each pair will present the completed phase in a Jupyter Notebook to their instructor during the practical session. State the percentage contribution of each member of the pair in the notebook. The report must then be submitted electronically by **one member of the pair** via UIS by Sunday, **22 November 2026 at 23:59**.

Practical Session Activities: 10 points

The report is due in Week 12 of the semester. Each pair will present the completed phase in a Jupyter Notebook to their instructor during the practical session. State the percentage contribution of each member of the pair in the notebook. The report must then be submitted electronically by **one member of the pair** via UIS by Sunday, **06 December 2025 at 23:59**.